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Activity-9

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Series SP-D19: Rainfall

1) Look at the series

- Observations : n = 60
- cycle length : 12 months
- full cycle : 5
- Trend : Rising
- slope : approximately 0.588 mm / month

R-code

```
plot(x, main = "Monthly Rainfall", ylab = "Rainfall (mm)")
coef(lm(rain ~ t))
```

2) Seasonal pattern

```
dc <- decompose(x)
print(dc$figure)
which.max(dc$figure)
which.min(dc$figure)
diff(range(dc$figure))
```

3) Consistency

```
monthplot(x, main = "Seasonal Pattern")
```

The seasonal pattern is broadly consistent across the five cycles. with some variable due to random noise.

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4) Autocorrelation

```
d ← resid(lm(rain ~ t))
a ← acf(d, lag.max = 12, plot = FALSE)$acf
a[13]
a[7]
2/sqrt(n)
```

lag 12 indicates annual seasonality if significantly positive.

lag 6 may be negative.

5) Model and advice

```
y1 ← diff(range(rain[1:12]))
y5 ← diff(range(rain[49:60]))
y5/y1
dc ← decompose(x, type = "additive")
adjusted ← x − dc$seasonal
coef(lm(as.numeric(adjusted) ~ t))
```

Model: Additive, because the seasonal amplitude is fixed.

Advice: Plan reservoir storage using the annual rainfall cycle and rising trend

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